

Engineering Electromagnetics II
Homework #4
48 points total

ECE 332 – Spring 2026

Due Friday, June 5, 2026

1. A 1-m long dipole is excited by a 1 MHz current with an amplitude of 12 A. What is the average power density radiated by the dipole at a distance of 5 km in a direction that is 45° from the dipole axis? (8 points)
2. The normalized radiation intensity of a certain antenna is given by

$$F(\theta) = \exp(-20\theta^2) \text{ for } 0 \leq \theta \leq \pi,$$

where θ is in radians. Determine: (15 points total)

- the half power beamwidth, (5 points)
 - the pattern solid angle, (5 points)
 - the antenna directivity. (5 points)
3. For a dipole antenna of length $l = \lambda/4$, (15 points total)
 - determine the directions of maximum radiation, (5 points)
 - obtain an expression for $S_{\max}$, (5 points)
 - generate a plot of the normalized radiation pattern $F(\theta)$. (5 points)

4. Determine the effective area of a half-wave dipole antenna at 100 MHz, and compare it with its physical cross section if the wire diameter is 2 cm. (5 points)
5. A half-wave dipole TV broadcast antenna transmits 1 kW at 50 MHz. What is the power received by a home television antenna with a 3 dB gain if located at a distance of 30 km? (5 points)